MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

ENCODER, FAULT ISOLATION, KEY CODE MFG

This amendment forms a part of Military Specification MIL-M-28787/344(EC), dated 18 April 1983, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

TABLE I: Delete and substitute new table I as printed on page 2 of this amendment.

PAGE 6

4.5.2, fourth line: Add "and test condition J" between the words "column" and "shall".

PAGE 8

FIGURE 2: Delete and substitute new figure 2 as printed on page 3 of this amendment.

PAGE 9

TABLE II: Delete and substitute new table II as printed on page 4 of this amendment.

PAGE 14

TABLE IV, test conditions D1 and D2, Min column: Delete "2.5" (two places) and substitute "2.50" (two places).

TABLE IV, test conditions D1 and D2, Max column: Delete "4.5" (two places) and substitute "4.55" (two places).

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FIGURE 8: Delete and substitute new figure 8 as printed on page 5 of this amendment.

PAGES 24 AND 25

FIGURE 9: Delete and substitute new figure 9 as printed on pages 6 and 7 of this amendment.

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FIGURE 15, note 4: Delete and substitute:

"4. Adjust item P07-2 as follows:

0.7 V ± 1% for test conditions P1-P20 and P25-P44.
2.5 V ± 1% for test conditions P21-P24 and P45-P48."

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FIGURE 18, note 2, first line: Delete "16" and substitute "11.8".

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(Project 5963-N616)

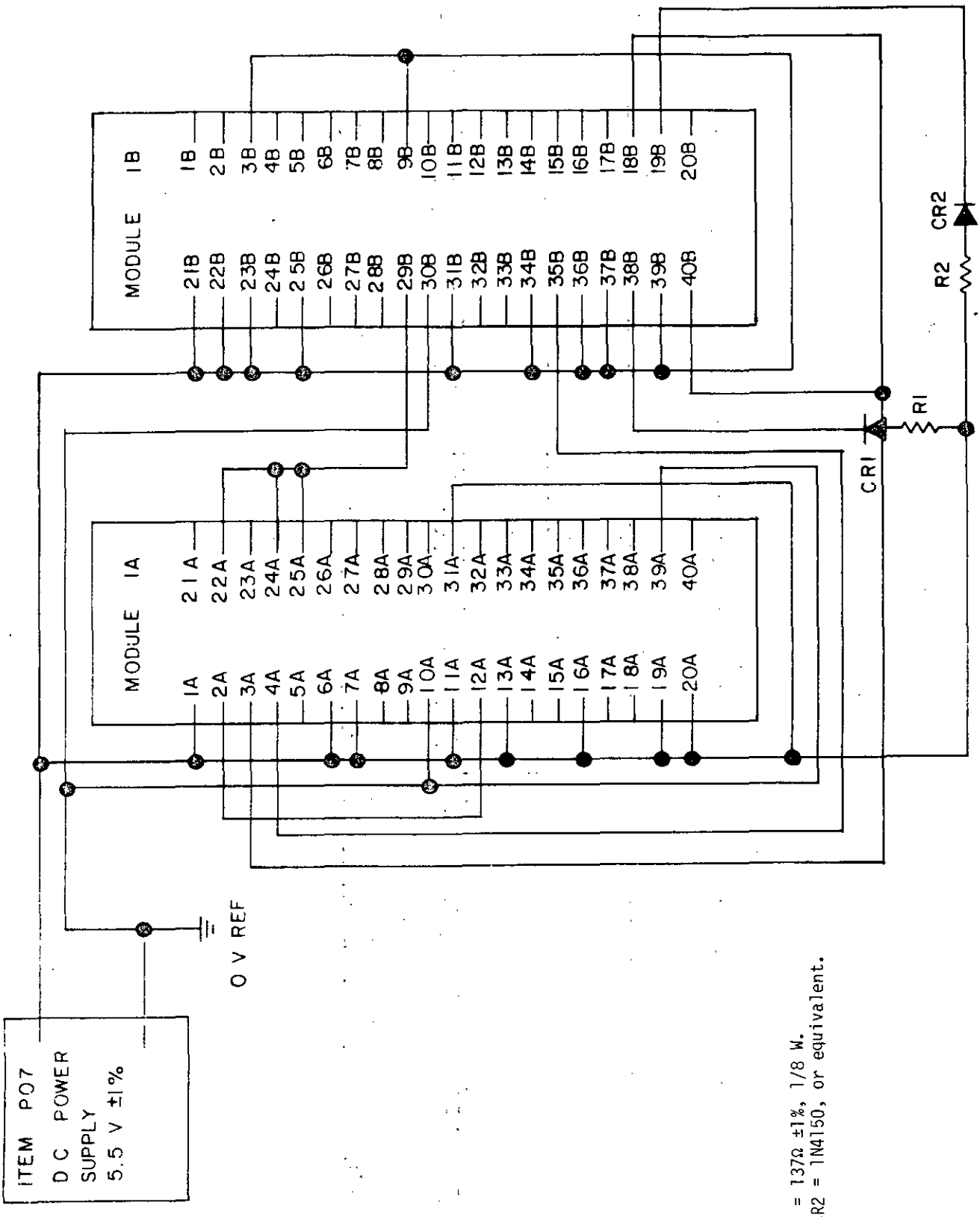
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FSC 5963

TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation	II	S	10	∞	1	∞	10	∞	1	∞	MΩ
A3-A16	Isolation	II	Q	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B4	Resistance	III	X	2024	2376	1914	2574	2024	2376	1914	2574	Ω
C1,C2	Resistance	III	X	138.0	162.0	130.5	169.5	138.0	162.0	130.5	169.5	Ω
D1,D2	NOR gate operation	IV	X	2.50	4.55	2.50	4.55	2.50	4.55	2.50	4.55	V
D5,D6,D10 D11	NOR gate operation	IV	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
D3,D4, D7-D9,D12	NOR gate operation	IV	X	0	400	0	400	0	400	0	400	mV
E1,E11	Exclusive OR/8 INPUT NAND gate operation (module 1A)	V	X	0	400	0	400	0	400	0	400	mV
E2-E10, E12-E19	Exclusive OR/8 INPUT NAND gate operation (module 1A)	V	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
F1,F11	Exclusive OR/8 INPUT NAND gate operation (module 1B)	VI	X	0	400	0	400	0	400	0	400	mV
F2-F10, F12-F19	Exclusive OR/8 INPUT NAND gate operation (module 1B)	VI	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
G1-G3,G5-G7	Independent NAND gate operation	VII	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
G4,G8	Independent NAND gate operation	VII	X	0	400	0	400	0	400	0	400	mV
H1-H5, H11-H13,H18	Encoder fault isolation	VIII	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
H6-H10, H14-H17	Encoder fault isolation	VIII	X	0	400	0	400	0	400	0	400	mV
J	Filter capacitance	IX	S	8	12	7	13	7	13	6	14	μF
K	Power supply current (V_{CC} max)	X	X	10	80	10	80	10	80	10	80	mA

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NOTES:
1. R1 = R2 = 137Ω ±1%, 1/8 W.
2. CR1 = CR2 = 1N4150, or equivalent.

FIGURE 2. Operating temperature, component temperature, and life test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	Frame	All pins	10	"	MΩ	3
A2	Isolation, used to unused pins	1A-7A,10A-13A, 15A,16A,19A, 20A,22A-25A, 31A,37A,39A, 40A,1B,3B,9B, 10B,17B-23B, 25B,29B-31B, 34B-40B	8A,9A,14A,17A, 18A,21A,26A-30A,32A-36A, 38A,2B,4B-8B, 11B-16B,24B, 26B-28B,32B, 33B	"	"	"	3
A3	Isolation, unused pins	8A	9A,14A,17A, 18A,21A,26A-30A,32A-36A, 38A	"	"	"	4
A4	"	9A	14A,17A,18A, 21A,26A-30A, 32A-36A,38A	"	"	"	"
A5	"	14A	17A,18A,21A, 26A-30A,32A-36A,38A	"	"	"	"
A6	"	17A	18A,21A,26A-30A,32A-36A, 38A	"	"	"	"
A7	"	18A	21A,26A-30A, 32A-36A,38A	"	"	"	"
A8	"	21A	26A-30A,32A-36A,38A	"	"	"	"
A9	"	26A-30A	32A-36A,38A	"	"	"	"
A10	"	32A-36A	38A	"	"	"	"
A11	"	2B	4B-8B,11B-16B,24B,26B-28B,32B,33B	"	"	"	"
A12	"	4B-8B	11B-16B,24B, 26B-28B,32B, 33B	"	"	"	"

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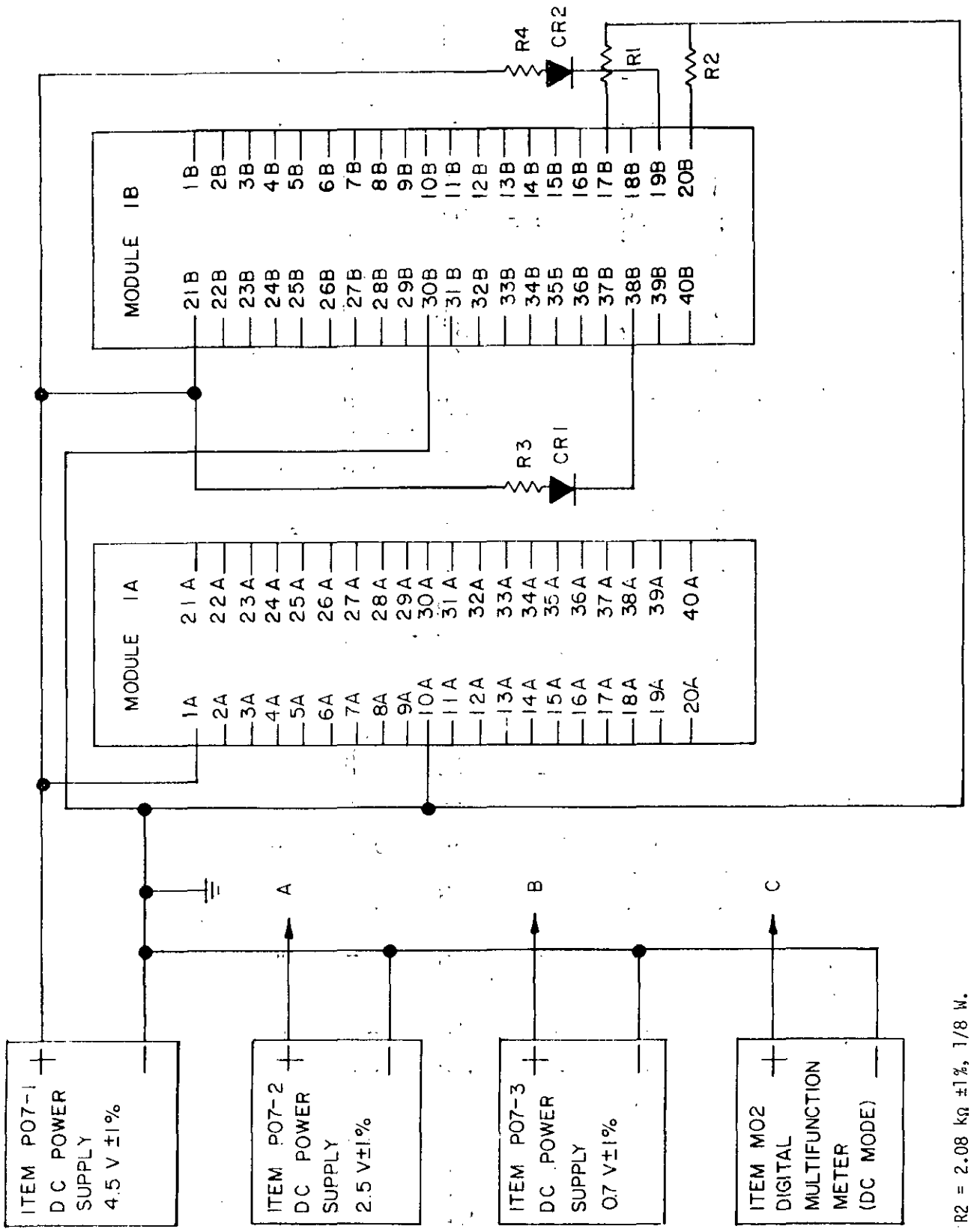


FIGURE 8. Independent NAND gate operation test circuit for table VII.

NOTES:
1. R1 = R2 = 2.08 kΩ ±1%, 1/8 W.
2. R3 = R4 = 97.6Ω ±1%, 1/8 W.
3. CR1 = CR2 = 1N4150, or equivalent.

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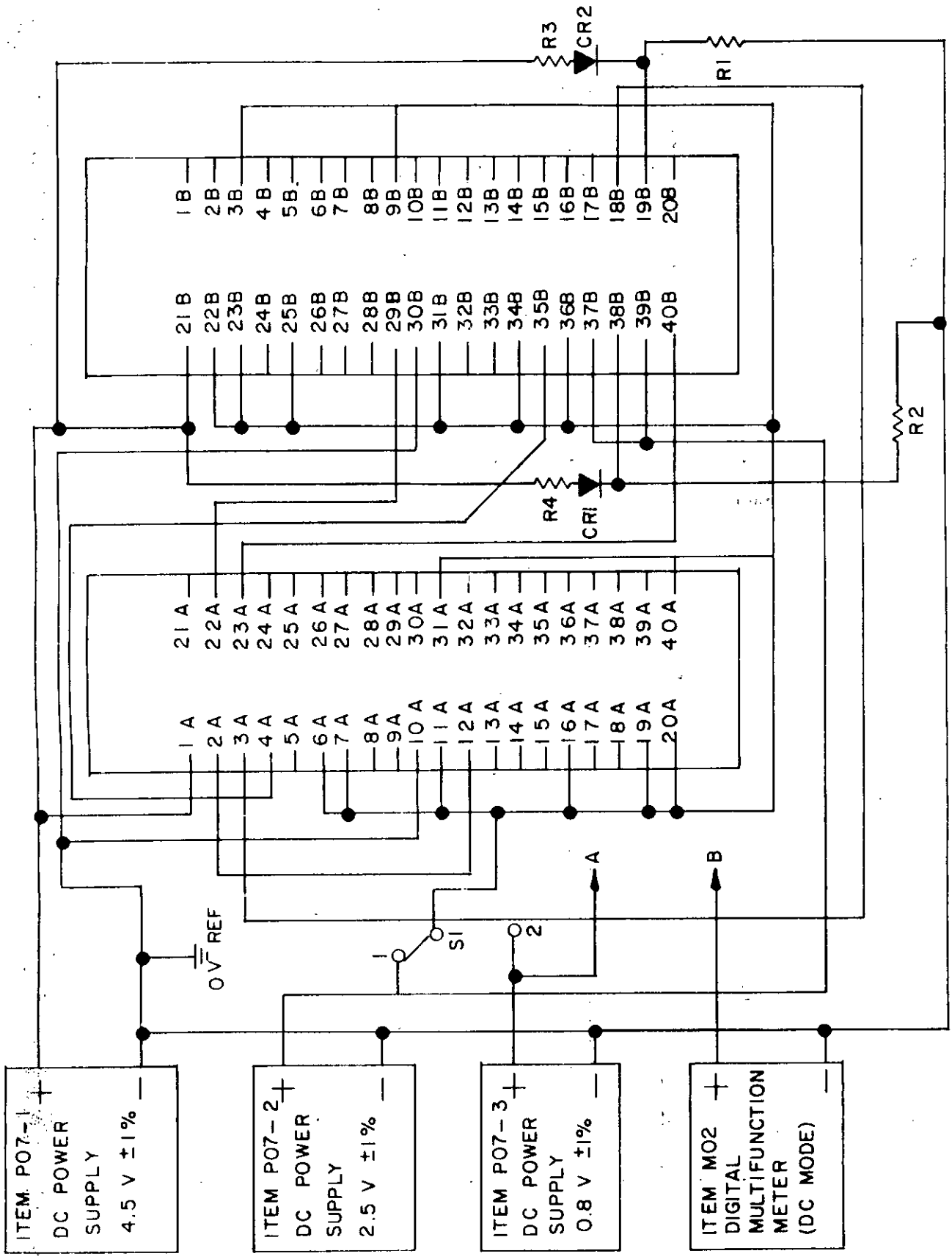


FIGURE 9. Encoder fault isolation test circuit for table VIII.

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NOTES:
1. R1 = R2 = 2.08 kΩ ±1%, 1/8 W.
2. R3 = R4 = 69.8Ω ±1%, 1/8 W.
3. S1 = SPDT switch.
4. CR1 = CR2 = 1N4150, or equivalent.
5. Set switch S1 to position 1 for test conditions H1, H3-H6, and H8-H10 and to position 2 for test conditions H2, H7, and H11-H18.

FIGURE 9. Encoder fault isolation test circuit for table VIII - Continued.

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